

Module 3 - JavaScript in HTML

1. How can JavaScript be added to an HTML page?

JavaScript can be added to an HTML page in three ways: Inline JavaScript, Internal JavaScript, and External JavaScript.

2. What is Inline JavaScript?

Inline JavaScript is JavaScript code written directly inside an HTML element using an event attribute.

Example:

```
<button onclick="alert('Hello!')">Click Me</button>
```

3. What is Internal JavaScript?

Internal JavaScript is JavaScript code written inside the `<script>` tag within the `<head>` or `<body>` section of an HTML page.

Example:

```
<script>
    document.write("Welcome to JavaScript!");
</script>
```

4. What is External JavaScript?

External JavaScript is JavaScript code written in a separate .js file and linked to an HTML page using the `<script>` tag with the `src` attribute.

Example:

```
<script src="index.js"></script>
```

5. Where should the `<script>` tag be placed for better performance?

The `<script>` tag is generally placed at the end of the `<body>` tag for better performance.

6. What is the use of the `src` attribute in the `<script>` tag?

The `src` attribute is used to specify the path or location of an external JavaScript file linked to an HTML page.

7. What is the difference between Inline, Internal, and External JavaScript?

Method	Where Written	Use	Pros	Cons
Inline JavaScript	Inside HTML element	Small actions (events)	Quick and simple	Not scalable, hard to maintain
Internal JavaScript	Inside <script> in HTML	Single-page scripts	No extra file needed	Not reusable across pages
External JavaScript	Separate .js file (src)	Large projects	Reusable, maintainable, cacheable	Needs extra HTTP request

8. What are the basic syntax rules of JavaScript?

- JavaScript is case-sensitive.
- JavaScript statements can be ended with a semicolon (;).
- Curly braces ({ }) are used to define code blocks.
- Indentation is used to improve code readability.
- Whitespace and line breaks are generally ignored, except inside strings.

Example Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>JavaScript in HTML</title>

  <!-- Internal JavaScript -->
  <script>
    console.log("Hello from Head!");
  </script>
</head>
<body>

  <h1>I'm gonna make you dynamic</h1>

  <!-- Inline JavaScript -->
  <button onclick="alert('Hello JavaScript!')">
    Click Me
  </button>
```

```
<!-- Internal JavaScript -->

<script>
  setTimeout(() => {
    document.querySelector("h1").innerHTML = "I'm Changed";
  }, 4000);

  console.log(100);
  console.log("JavaScript");

  document.write("JavaScript INTRO");
</script>

<!-- External JavaScript -->
<script src="index.js"></script>

</body>
</html>
```